PHYS30762 Object Oriented Programming in C++

Analogue Circuits Report

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Abstract

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1. Introduction

This program allows users to build analogue circuits using resistors, capacitors, and inductors (which are all classes derived from an abstract base `Component` class). A `Circuit` class can be set to be series or parallel, and any number of components can be added to it, which is then stores as a vector of components (specifically a vector of unique pointers to components). Since the `Circuit` class is inherited from the `Component` class itself, it is possible to add series circuits to parallel circuits, or vice versa, and therefore construct circuits of any topology.

In the main file of this program, a *library* of components is created, which stores different types of resistors, capacitors and inductors. These components are validated through the use of validation files, such that if the user tries to initialise a component with properties which do not match those specified in the validation file, an error message will be returned. The validation files specify a component name, like *Wirewound* for resistor, as well as a tolerance to dissipative effects, and a maximum frequency to which they can withstand [5]. Each validation file then contains maximum and minimum possible values for the resistance, capacitance and inductance respectively.

The `Component` base class contains an operating frequency `frequency` and a unique ID `component_id`, as well as a function to calculate the impedance of the component (using the complex library in C++), and its magnitude and phase difference (`get_impedance()`, `get_impedance_magnitude`, `get_phase_difference`). It also has a `base_info()` function which outputs these properties, and which is overriden in the derived classes to print the resistance, capacitance and inductance of the corresponding classes. A summary of the data members and member functions can be seen in figure X.

This program therefore gives the user an easy way of constructing circuits with nested branches and varying topologies, as well as ensuring at the same time that the components are physically meaningful, much like an undergraduate circuit labaratory where only certain types of components are available and can be used to build circuits.

2. Code design and implementation

We continue with this idea of the program being a simulation of a circuits lab so that the logic behind it can be more clearly explained. Consider the user of the program to be a student in an undergraduate circuits lab. This student wants to build a circuit using components from the lab's demonstrator, who has this library of components available. This library must contain components which are manufacturable, and we must have some way of enforcing this. The validation files can be seen as catalogues of manafacturable components from which the lab demonstrator must choose from to add to their library.

The base `Component` class contains two functions which are used by the derived classes to perform the validation checks on the components. The first is `load_validation_file` which takes in the name of the file, as well as the record to which to save it to. This record is something each derived class has, and is a vector of the `ValidationRecord` struct, whose data members are equivalent to each field in a line of the validation files (component name, maximum value, minimum value, tolerance, maximum frequency). `load_validation_file`also performs checks on the file for any blank lines or extra characters. The second function which `Component` has is `validate_component`. This function takes values from a user who wishes to intialise a new component or circuit, and

checks that they match the record for the corresponding class. `validate_component` is used within all constructors and setters of the derived classes, apart from the `Circuit` class which does not have any validation checks. That being said, the `Circuit` class does use `validate_component` if a component is added to a circuit, or its frequency if set via a setter, to check that the frequency being set does not violate the maximum frequency of a component inside a circuit. This is because when a component is added to a circuit, its frequency is set to match the circuit's frequency. In the analogy, it can be thought that the lab demonstrator is keeping the components at some set frequency in their library, which is changed when they are added to a circuit.

Within the validation function, it was useful to use a lambda function because checks for extra characters and blank lines were the same for all fields in the file. Listing 1 shows the lambda function used to perform checks for extra characters in each field in one line of the file. Line 1 is where the lambda function is declared, and the entries in the square brackets is where its use becomes really beneficial - `line` and `comonent_type_str` are passed into the lambda function *implicitly* without needing to be passed as a parameter. This means the function can take the current line which is being read, and the component type (e.g. resistor) from the lines above the lambda function (which is already inside the validation function). These are useful in producing custom error messages on lines 8 and 9, making it easier for the user to understand where something exactly has gone wrong - it prints out the component name so we know which validation file the problem has occurred, and it prints out the whole line so we know where to search for the problem in that file.

```
1     auto parse_double = [&line, &component_type_str](const
    string& field, const string& field_name, double& output)
2     {
3       std::stringstream field_ss(field);
4       string extra;
5
6       if(!(field_ss >> output) || (field_ss>>extra))
7       {
8         std::cerr<<component_type_str<<" validation file
    error: Extra or invalid characters in "<<
9           field_name<<" in the line {"<<line<<"}. Line
    will not contribute."<<std::endl;
10
11        return false; //parse failed
12      }
13
14      return true;
```

Listing 1: Lambda function in validate component function (C++)

If the student wants to use one of the components from the library of the lab demonstrator, they cannot take it directly away from the lab demonstrator, as they are the owner of that specific component. This follows from the fact that the the library which is set up in main is a vector of unique pointers. There was a consideration to make it a vector of shared pointers, in which case the student could "share" the component with the lab demonstrator, and put it in the circuit (which would also share ownership). This came with a big drawback, which was that if a component is added to a circuit, its frequency would be updated, and since its shared, this would affect the component in all places it was used. Listing 2 shows the case where the component is shared amongst two circuits,

and when the frequency of the second one is set to a different value of the first, and then we try to access the info of the first circuit, the component's frequency will not match the circuit's frequency, giving the incorrect calculation of impedance for that circuit.

To avoid this, the student must make a copy of the component from the library, at which point they can add it to their circuit. This required the introduction of a `clone()` function which all of the `Component`-derived classes were given. It performs a deep copy of the component, akin to how a copy constructor works. This also came to be very useful for the copy constructor and assignment operator in the Circuit class, which containing a vector of unique pointers must obide by at least the rule of 3. Since we do want the user to also be able to move the circuit from one place to another, we implement the rule of 5, but set the move copy constructor and assignment operator to be default, as it is correctly handled by the unique pointer's `std::move` function.

```
circuit1.setFrequency(50);
circuit2.setFrequency(1000);  // overwrites the component's
   frequency
circuit1.info();              // component now thinks f=1000
```

Listing 2: Shared pointers don't work well for library of components

If they let the student take that one component which they own, then no one else in the lab could use it. In that sense it is not logical for identical components to be stored twice in the library. The student must make a copy of the component from the library, at which point they can add it to their circuit. This is done via a `clone()` function which all of the `Component`-derived classes have. They could initialise a component just by itself and then move it to the circuit, but its easier to just clone inside the circuit infrastructure.

One must be careful about the copy semantics for the component classes when it comes to the IDs of each component.

3. Results

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4. Discussion and conclusions

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